

Research Areas

- Strongly correlated quantum matter
- Excitonic insulators and condensates
- Two-dimensional materials
- Low-temperature optics and transport

Academic Appointments

2026 – **Stanford Science Fellow**, Stanford University

Incoming fellow starting July 2026

Faculty host: Prof. Tony F. Heinz

Education

2020 – 2026 **Ph.D. in physics**, University of California, Berkeley

Advisor: Prof. Feng Wang

Thesis: Correlated exciton physics in van der Waals electron-hole bilayers

2020 – 2022 **M.A. in physics**, University of California, Berkeley

2016 – 2020 **B.S. in physics**, Peking University (PKU)

GPA: 3.9/4; Rank: 1/200⁺

Graduation honors: Weiming Scholar; Outstanding Graduate in Peking University; Outstanding Graduate of Beijing

Publications

30⁺ peer-reviewed articles; 1000⁺ citations. Full publication record: [🔍 Google Scholar](#).

†: equal contribution; ‡: corresponding author.

Selected first- and corresponding-author publications

1. **R. Qi**^{†‡}, Q. Li[†], J. Nie, ..., F. Wang[‡]. Observation of two-component exciton condensates in an excitonic insulator. **Nature** accepted. arXiv:2603.15443 (2026). [🔗](#)
2. **R. Qi**^{†‡}, Q. Li[†], Z. Zhang, ..., F. Wang[‡]. Gate-tunable biexcitons in electron-hole-electron trilayers. Under revision at **Science**.
3. **R. Qi**[†], Q. Li[†], H. Kim, ..., F. Wang[‡]. An exciton crystal in a moiré excitonic insulator. **Nature Physics** 22, 514-520 (2026). [🔗](#)
Featured in *Nature Physics* [News & Views](#); *Nature Reviews Materials* [Research Highlight](#)
4. **R. Qi**[†], Q. Li[†], Z. Zhang, ..., F. Wang[‡]. Competition between excitonic insulators and quantum Hall states in correlated electron-hole bilayers. **Nature Materials** 25, 35-41 (2026). [🔗](#)
5. Z. Zhang^{†‡}, **R. Qi**[†], J. Xie, ..., F. Wang[‡]. Orbital dependent Coulomb drag in electron-hole bilayer graphene heterostructures. **Physical Review Letters** 136, 126303 (2026). [🔗](#)
Featured in *Physical Review Letters* [Editor's suggestion](#)
6. Z. Lu^{†‡}, **R. Qi**[†], R. Dutta[†], ..., F. Wang[‡]. Nature of emergent moiré excitations in

MoSe₂/WS₂ moiré superlattices. **Nano Letters** 26 (12), 4096-4102 (2026). [🔗](#)

7. **R. Qi**, Q. Li, Z. Zhang, ..., F. Wang[‡]. Electrically controlled interlayer trion fluid in electron-hole bilayers. **Science** 390 (6770), 299-303 (2025). [🔗](#)
Featured in *Science Perspective*
8. **R. Qi**[†], A. Joe^{‡‡}, Z. Zhang, ..., F. Wang[‡]. Perfect Coulomb drag and exciton transport in an excitonic insulator. **Science** 388 (6744), 278-283 (2025). [🔗](#)
Featured in *Journal Club for Condensed Matter Physics*
9. **R. Qi**[†], A. Joe^{‡‡}, Z. Zhang, ..., F. Wang[‡]. Thermodynamic behavior of correlated electron-hole fluids in van der Waals heterostructures. **Nature Communications** 14, 8264 (2023). [🔗](#)
10. **R. Qi**[†], R. Shi[†], Y. Li, ..., P. Gao[‡]. Measuring phonon dispersion at an interface. **Nature** 599, 399-403 (2021). [🔗](#)
11. **R. Qi**[†], N. Li[†], J. Du, ..., P. Gao[‡]. Four-dimensional vibrational spectroscopy for nanoscale mapping of phonon dispersion in BN nanotubes. **Nature Communications** 12, 1179 (2021). [🔗](#)
12. **R. Qi**[†], R. Wang[†], Y. Li, ..., P. Gao[‡]. Probing far-infrared surface phonon polaritons in semiconductor nanostructures at nanoscale. **Nano Letters** 19 (8), 5070-5076 (2019). [🔗](#)

Honors and Fellowships

- 2026 – 2029 Stanford Science Fellow, Stanford University [🔗](#)
2023 – 2024 Kavli ENSI Graduate Student Fellow, Kavli Energy NanoScience Institute [🔗](#)
2018, 2019 National Scholarship (China)

Research Experience

- 2021 – 2026 **Graduate Student Researcher**, Department of Physics, UC Berkeley
Ultrafast nano-optics group (PI: Prof. Feng Wang)
- Established electrically tunable van der Waals electron-hole bilayers as a platform for correlated exciton phases, including interlayer excitons, trions, biexcitons, exciton condensates, and exciton crystals.
 - Developed optical spectroscopy approaches for measuring compressibility, Coulomb drag, and transport properties in strongly interacting two-dimensional systems.
 - Designed and performed low-temperature, high-magnetic-field optical and transport experiments down to 10 mK and up to 12 T.
- 2018 – 2021 **Undergraduate Research Assistant**, Peking University
Electron microscopy laboratory (PI: Prof. Peng Gao)
- Developed electron energy-loss spectroscopy techniques combining nanometer spatial resolution, high momentum resolution, and milli-electron-volt energy resolution.
 - Investigated local lattice dynamics at heterointerfaces and nanostructures.

Teaching and Mentoring

- 2020 – 2021 **Graduate Student Instructor**, Department of Physics, UC Berkeley
Physics 7A: Physics for Scientists and Engineers (Fall 2020, Spring 2021)
Led discussion and laboratory sections; graded homework and exams; held office hours.

- 2023 – 2024 **Undergraduate Research Mentor**, Berkeley Physics Innovators Initiative (Pi²)
Mentored Berkeley undergraduate researchers (selective funded research program).
- 2022, 2025 **Undergraduate Research Mentor**, Berkeley Physics International Education (BPIE)
Mentored visiting undergraduate researchers.

Talks and Presentations

- 2026 “Correlated exciton physics in strongly coupled electron-hole bilayers”, APS March Meeting, Denver [🔗](#)
- 2026 “Excitonic insulators and condensates in electron-hole bilayers” (Poster), New Frontiers in Nanoscale Materials 2026 Symposium
- 2025 “Interlayer excitons and trions in electron-hole bilayers”, Berkeley Condensed Matter Seminar [🔗](#)
- 2024 “Perfect Coulomb drag and exciton transport in an excitonic insulator”, APS March Meeting, Minneapolis [🔗](#)
- 2024 “Correlated interlayer excitons and trions in electron-hole bilayers” (Poster), Kavli ENSI 10th Anniversary Symposium [🔗](#)